



School of Computer Science and Information Technology

Department of BCA

I Semester - Specialization(Section)

22BCA3C01: FUNDAMENTALS OF MATHEMATICS

Activity 1

Power Point Presentation

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CERTIFICATE

This is to certify that Mr. **Bharath, Lakshith, Lochan, Ravi Tej, Rohith** has satisfactorily completed activity prescribed by JAIN Deemed to be university for the fourth semester degree course in the year 2023-2024.

Assignment topic (**Set Theory**) for Activity-1

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Declaration

I hereby declare that the work which is being prepared in this activity report "Set Theory" for the award of the degree of Bachelor of computer Application ,submitted in the Department of Bachelor of computer Application Jain Deemed to be University ,Jayanagar 560068 ,Bangalore,Karnataka, India Under the Supervision of Lavanya D ,Assistant Professor.

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Set Theory

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Introduction

Introduction

Set Theory is a fundamental branch of mathematics that deals with sets, which are collections of objects.

Purpose

This presentation aims to provide a clear understanding of key Set Theory concepts and their practical applications.

Scope

We will cover set operations, subsets, counting, Venn diagrams, relations, functions, the Pigeonhole Principle, and problem-solving examples to equip you with a solid foundation in Set Theory.

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Set Operations

Define Sets:

A set is a collection of distinct objects or elements grouped together as a single entity. Sets are used to represent and organize data in mathematics.

Define Elements:

Elements, also known as members or items, are the individual objects that belong to a set. Each element within a set is unique and contributes to the set's identity.

Explain Set Notation: Set notation is a concise way to represent sets using braces {} and commas to list its elements. For example:

Set A = {1, 2, 3, 4, 5} represents a set A with elements 1, 2, 3, 4, and 5.

Set B = {apple, banana, cherry} represents a set B with elements "apple," "banana," and "cherry."

Set notation allows us to define sets, manipulate them with set operations, and work with mathematical expressions involving sets efficiently.

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Common Set Operations

- **Union ($\cup$):** The union of two sets, A and B (denoted as $A \cup B$), is a new set that contains all unique elements from both A and B.
- **Intersection ($\cap$):** The intersection of two sets, A and B (denoted as $A \cap B$), is a new set that contains only the elements that are common to both A and B.
- **Complement ($\neg$ or $'$):** The complement of a set A (denoted as $\neg A$ or A') contains all elements not present in set A within a universal set U.
- **Difference ($-$ or $\setminus$):** The difference of sets A and B (denoted as $A - B$ or $A \setminus B$) contains all elements that are in A but not in B.
- **Cartesian Product ($\times$):** The Cartesian product of sets A and B (denoted as $A \times B$) is a set of ordered pairs, where each element of A is paired with each element of B.

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Examples of Set Operations

Example 1:

Union ($\cup$)

Set A = {1, 2, 3}

Set B = {3, 4, 5}

$A \cup B = \{1, 2, 3, 4, 5\}$

Example 2:

Intersection ($\cap$)

Set A = {apple, banana, cherry}

Set B = {banana, cherry, grape}

$A \cap B = \{\text{banana, cherry}\}$

Example 3:

Complement ($\neg A$)

Universal Set U = {1, 2, 3, 4, 5}

Set A = {2, 4}

$\neg A$ or $A' = \{1, 3, 5\}$

Example 4:

Difference ($A - B$)

Set A = {1, 2, 3, 4, 5}

Set B = {3, 4}

$A - B$ or $A \setminus B = \{1, 2, 5\}$

Example 5:

Cartesian Product ($A \times B$)

Set A = {x, y}

Set B = {1, 2}

$A \times B = \{(x, 1), (x, 2), (y, 1), (y, 2)\}$

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Subsets

Define Subsets:

A subset is a set that contains only elements that are also found in another set. In other words, set A is a subset of set B (denoted as $A \subseteq B$) if every element in A is also an element in B.

Define Proper Subsets:

A proper subset is a subset that contains some, but not all, of the elements of another set. Set A is a proper subset of set B (denoted as $A \subset B$) if every element in A is in B, and A is not equal to B.

How to Determine Subsets:

To determine if set A is a subset of set B, check if every element of A is also in B. If this condition holds, then A is a subset of B.

For proper subsets, ensure that A is a subset of B ($A \subseteq B$), but A is not equal to B ($A \neq B$).

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Examples of Subsets

Example 1: Subset ($A \subseteq B$)

Set A = {1, 2, 3}

Set B = {1, 2, 3, 4, 5}

A is a subset of B because all elements in A (1, 2, and 3) are also in B.

Example 2: Proper Subset ($A \subset B$)

Set A = {red, green}

Set B = {red, green, blue}

A is a proper subset of B because all elements in A (red and green) are in B, but A is not equal to B.

Example 3: Not a Subset

Set A = {a, b, c}

Set B = {x, y, z}

A is not a subset of B because none of the elements in A are in B.

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Counting and Venn Diagrams

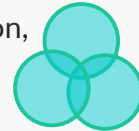
Use of Venn Diagrams:

Venn diagrams are graphical tools used to visually represent sets and their relationships.

They consist of overlapping circles, with each circle representing a set.

Overlapping regions show elements shared between sets, while non-overlapping regions represent unique elements.

Venn diagrams help in understanding set operations like union, intersection, and complement.



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Principle of Inclusion-Exclusion

Explaining the Principle of Inclusion-Exclusion:

- The Principle of Inclusion-Exclusion is a valuable counting technique in combinatorics and set theory.
- It allows us to calculate the size of the union of multiple sets accurately, taking into account the shared elements.
- This principle is particularly useful when dealing with sets that have overlapping elements, such as events in probability or items in a survey.

How It Works:

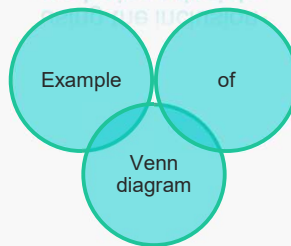
- The basic idea is to count the elements in the union of sets while making sure that no element is counted more than once.
- The principle uses a systematic approach to add and subtract the sizes of various intersections of sets.

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Examples of Counting and Venn Diagrams

Sets $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $C = \{3, 4, 5\}$
Display a Venn diagram illustrating the relationships between these sets, highlighting the overlapping regions.

Sets $A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$, $C = \{5, 6, 7\}$
Calculate the size of the union of these sets using the inclusion-exclusion principle.



Set $U = \{1, 2, 3, 4, 5, 6\}$
Sets $A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$, $C = \{5, 6\}$
Use a Venn diagram to count the number of elements in $A \cup B \cup C$.

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Relation

Defining Relations in Set Theory:

In set theory, a relation between sets A and B is defined as a subset of the Cartesian product of A and B . In other words, it's a collection of ordered pairs (a, b) , where ' a ' is an element of set A , and ' b ' is an element of set B .

Relations can be used to model connections, interactions, or associations between elements of different sets.

Relations are represented as $R \subseteq A \times B$, where R is the relation between sets A and B .

Applications:

Relations have extensive applications in various fields, including mathematics, computer science, and social sciences.

They are used in graph theory to model networks, in database systems for defining relationships between database tables, and in formal logic for representing logical implications.

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Different Types of Relations

Reflexive Relations:

A relation R on a set A is reflexive if, for every element a in A , (a, a) is in R .

In simpler terms, it means that every element in the set is related to itself.

Example: The relation "is equal to" is reflexive on the set of real numbers.

Symmetric Relations:

A relation R on a set A is symmetric if, for every pair (a, b) in R , (b, a) is also in R .

In other words, if a is related to b , then b is related to a .

Example: The relation "is a sibling of" is symmetric.

Transitive Relations:

A relation R on a set A is transitive if, for every pair (a, b) and (b, c) in R , (a, c) is also in R .

This means that if a is related to b , and b is related to c , then a is related to c .

Example: The relation "is less than" is transitive on the set of integers.

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Examples of Relations

- Set $A = \{1, 2, 3\}$
- Relation $R = \{(1, 1), (2, 2), (3, 3)\}$
- R is reflexive because every element in A is related to itself.

*Reflexive
Relation*

- Set $B = \{\text{red, green, blue}\}$
- Relation $S = \{(\text{red, green}), (\text{green, red}), (\text{blue, blue})\}$
- S is symmetric because for every pair (a, b) in S , (b, a) is also in S .

*Symmetric
Relation*

- Set $C = \{A, B, C\}$
- Relation $T = \{(A, B), (B, C), (A, C)\}$
- T is transitive because for every pair (a, b) and (b, c) in T , (a, c) is also in T .

*Transitive
Relation*



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Functions

Defining Functions in Set Theory:

Functions, also known as mappings or transformations, play a fundamental role in mathematics.

In set theory, a function is a rule or relation that assigns each element in the domain to exactly one element in the codomain.

Functions can be thought of as processes that take inputs and produce outputs, allowing us to model a wide range of real-world phenomena.

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Types of Functions

Plain Functions (Many-to-One):

Plain functions, also known as many-to-one functions, allow multiple elements from the domain to map to the same element in the codomain.

In other words, for a given input, there can be multiple outputs.

Example: The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$ is a plain function because, for any real number "x," there are multiple values of "x" that produce the same output (e.g., $f(2) = 4$ and $f(-2) = 4$).

One-to-One Functions (Injective Functions):

One-to-one functions, also called injective functions, ensure that no two distinct elements in the domain map to the same element in the codomain.

Each element in the domain has a unique mapping to an element in the codomain.

Example: The function $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = 2x$ is a one-to-one function because for any real numbers "x" and "y" where $x \neq y$, $g(x) \neq g(y)$.

Onto Functions (Surjective Functions):

Onto functions, or surjective functions, ensure that every element in the codomain has at least one element in the domain that maps to it.

The function covers the entire codomain.

Example: The function $h: \mathbb{R} \rightarrow \mathbb{R}$ defined by $h(x) = 3x$ is an onto function because every real number is covered as there is an "x" value for which $h(x)$ equals that number.

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Function Composition and Inverse Functions

Explaining Function Composition:

Function composition is a mathematical operation that combines two or more functions to create a new function.

When you compose functions, you apply one function to the result of another function, creating a chain of transformations.

Function composition is denoted as $(f \circ g)(x)$, where "f" and "g" are the functions being composed.

Introducing the Concept of Inverse Functions:

Inverse functions are functions that "undo" the effect of another function.

If a function "f" maps input "x" to output "y," its inverse function, denoted as " f^{-1} ," maps "y" back to "x."

Inverse functions allow you to reverse a process, making them essential for solving equations and modeling inverse relationships.

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Examples of Function Composition

Example 1: Function Composition

Functions: $f(x) = 2x$ and $g(x) = x + 3$

Find $(f \circ g)(x)$.

Explanation: $(f \circ g)(x) = f(g(x)) = f(x + 3) = 2(x + 3) = 2x + 6$.

Example 2: Function Composition

Functions: $h(x) = x^2$ and $k(x) = \sqrt{x}$

Find $(h \circ k)(x)$.

Explanation: $(h \circ k)(x) = h(k(x)) = h(\sqrt{x}) = (\sqrt{x})^2 = x$.

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Finding Inverse Functions

Finding the Inverse Function

Function: $f(x) = 3x - 2$

Find the inverse function $f^{-1}(x)$.

Steps:

Replace " $f(x)$ " with " y ."

Swap " x " and " y " to solve for " y ."

Replace " y " with " $f^{-1}(x)$ " to find the inverse function.

Calculation:

$$y = 3x - 2$$

Swap " x " and " y ": $x = 3y - 2$

Solve for " y ": $3y = x + 2$

Divide by 3: $y = (x + 2)/3$

Replace " y " with " $f^{-1}(x)$ ": $f^{-1}(x) = (x + 2)/3$

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The Pigeonhole Principle

Defining the Pigeonhole Principle:

The Pigeonhole Principle, also known as Dirichlet's Box Principle, is a fundamental principle in combinatorics and mathematics.

It states that if you try to distribute more items into a set of containers (pigeonholes) than there are containers, at least one container must contain more than one item.

In simpler terms, if you have too many objects to place into fewer categories, some categories will necessarily have more than one object.

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Examples and Applications

Example 1: Birthday Paradox

Example: In a group of just 23 people, there's a better than even chance that at least two people share the same birthday.

Explanation: Even though there are 365 possible birthdays, the Pigeonhole Principle shows that with 23 people, you have 23 "pigeons" (people) to place in 365 "pigeonholes" (birthdays), making it highly likely that at least one day will have more than one person.

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Example 2: Sock Color Matching

Example: In a dark room, you have 11 red socks and 10 blue socks. How many socks do you need to pull out to guarantee a matching pair?

Explanation: According to the Pigeonhole Principle, you would need to pull out at least 12 socks to guarantee a matching pair because there are 11 pigeonholes (red socks) and 12 pigeons (socks pulled out).

Example 3: Telephone Directory

Example: In a telephone directory, there are 10,000 phone numbers with unique last four digits. How many numbers do you need to call to guarantee you reach someone with the same last four digits as your own?

Explanation: The Pigeonhole Principle tells us that you would need to call 10,001 numbers because there are 10,000 pigeonholes (unique last four digits) and 10,001 pigeons (numbers dialed).

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Relation to Set Theory

Explaining the Relation to Set Theory:

The Pigeonhole Principle is closely related to set theory and combinatorics.

In set theory, it's often used to prove the existence of certain combinatorial structures, such as repetitions, collisions, or matches.

It helps us understand and analyze situations where we need to distribute or categorize objects among sets or containers.

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Problems with Solutions

Problem 1: Set Theory

Given two sets $A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7, 8\}$, find the union of A and B ($A \cup B$).

Solution:

The union of sets A and B includes all unique elements from both sets.

$$A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$$

Problem 2: Relations

Determine whether the relation R on the set of integers is symmetric. Relation R: $(a, b) \in R$ if $a + b = 10$.

Solution:

A relation is symmetric if $(a, b) \in R$ implies $(b, a) \in R$ for all a, b in the set.

If $a + b = 10$, then $b + a = 10$, so the relation R is symmetric.

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Problem 3: Functions

Problem Statement: Find the composition of the following functions:

$$f(x) = 2x$$

$$g(x) = x - 1$$

Solution:

To find $(f \circ g)(x)$, we first apply g to x and then apply f to the result:

$$(f \circ g)(x) = f(g(x))$$

$$g(x) = x - 1$$

$$f(g(x)) = f(x - 1) = 2(x - 1) = 2x - 2$$

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Problem 4: The Pigeonhole Principle

Problem Statement:

In a deck of 52 playing cards, what is the minimum number of cards you need to draw to guarantee that you have at least two cards of the same suit (hearts, diamonds, clubs, or spades)?

Solution:

There are four suits, and the worst-case scenario is that you draw one card from each suit.

However, drawing a fifth card guarantees that it will belong to a suit you've already drawn.

So, you need to draw a minimum of 5 cards to guarantee having at least two cards of the same suit.

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Conclusion:

In this presentation, we explored key concepts in Set Theory, Relations, Functions, and the Pigeonhole Principle. Let's recap the key points:

Set Theory: We began by defining sets, elements, and set notation. We discussed common set operations such as union, intersection, and complement, which are fundamental for managing and analyzing data in various fields.

Relations: We defined relations as connections between sets and explored different types, including reflexive, symmetric, and transitive relations. Understanding relations helps us model relationships and make logical inferences.

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Functions: We introduced functions as mappings between sets, highlighting plain, one-to-one, and onto functions. Functions are essential for modeling and solving problems in mathematics and various scientific disciplines.

Pigeonhole Principle: We explained the Pigeonhole Principle, which asserts that if you distribute items into containers, some containers must contain more than one item. We explored examples and applications, emphasizing its importance in probability and combinatorics.

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Importance and Relevance of Set Theory:

Set Theory is a foundational branch of mathematics with wide-ranging applications. It provides a structured framework for organizing, analyzing, and solving problems in various fields, including:

Mathematics: Set Theory forms the basis for most mathematical structures and concepts, including numbers, algebra, and calculus.

Computer Science: Sets are fundamental in data structures, databases, and algorithms, allowing efficient data management and retrieval.

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Statistics: Set operations are essential for data analysis, sampling, and probability theory.

Engineering: Set Theory is used in optimization, control theory, and modeling complex systems.

Logic: Set Theory is intimately tied to logical reasoning and formal proofs.

Real-world Applications: From organizing data in spreadsheets to modeling complex systems in science and engineering, Set Theory is a practical tool in numerous professions.

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Reference:

- Content generated by ChatGPT, a language model developed by OpenAI, based on a wide range of general knowledge up to September 2021.
- "Introduction to Set Theory" by Karel Hrbacek and Thomas Jech
- "Discrete Mathematics and Its Applications" by Kenneth H. Rosen
- "Principles of Mathematical Analysis" by Walter Rudin (for topics related to functions)
- Various online educational resources, articles, and academic papers related to the Pigeonhole Principle and problem-solving techniques in mathematics.

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